

Power Series Solution: Legendre's Equation

The differential equation

$$\left[(1-x^2) \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} + n(n+1)y = 0 \right] \rightarrow \textcircled{1}$$

where n is real number, is called Legendre's differential equation.

Here $x=0$ is an ordinary point, take the solution as

$$y = \sum_{k=0}^{\infty} a_k x^k$$

And the solution of $\textcircled{1}$ is obtained as

$$\begin{aligned} y &= a_0 + a_1 x + a_2 x^2 + \dots \\ &= a_0 \left[1 - \frac{n(n+1)}{2!} x^2 + \frac{(n-2)n(n+1)(n+3)}{4!} x^4 - \dots \right] \\ &\quad + a_1 \left[x - \frac{(n-1)(n+2)}{3!} x^3 + \frac{(n-3)(n-1)(n+2)(n+4)}{5!} x^5 - \dots \right] \\ &= a_0 y_1(x) + a_1 y_2(x) \end{aligned}$$

where

$$y_1(x) = 1 + \sum_{k=1}^{\infty} (-1)^k \frac{n(n-2) \cdots (n-2k+2)(n+1)(n+3) \cdots (n+2k)}{(2n)!} x^{2k} \quad \rightarrow (2)$$

and

$$y_2(x) = x + \sum_{k=1}^{\infty} (-1)^k \frac{(n-1)(n-3) \cdots (n-2k+1)(n+2)(n+4) \cdots (n+2k)}{(2n+1)!} x^{2k+1} \quad \rightarrow (3)$$

Case I When n is even say $n = 2m$, from (2)

$$y_1(x) = 1 + \frac{(m!)^2}{(2m)!} \sum_{k=1}^{\infty} (-1)^k \frac{(2m+2k)!}{(m-k)!(m+k)!(2k)!} x^{2k},$$

$y_1(x)$ is a polynomial of degree $2m$, but $y_2(x)$ is not.

Case II When n is odd, say $n = 2m+1$, from (3),

$$y_2(x) = x + \frac{(m!)^2}{(2m+1)!} \sum_{k=1}^{\infty} (-1)^k \frac{(2m+2k+1)!}{(m-k)!(m+k)!(2k+1)!} x^{2k+1}$$

When n is odd $y_2(x)$ is a polynomial, but $y_1(x)$ is not.

§ The Legendre's polynomial (of 1st kind)

$$P_n(x) = \sum_{k=0}^n \frac{(-1)^k (2n-2k)!}{2^n \cdot k! (n-k)! (n-2k)!} x^{n-2k}$$

$$\text{where } n = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ \frac{n-1}{2}, & \text{if } n \text{ is odd.} \end{cases}$$

(2)

§ Rodrigue's formula (as Legendre's Polynomial)

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2-1)^n \rightarrow \textcircled{A}$$

Putting $n=0, 1, 2, 3, \dots$ we get polynomials $P_0(x), P_1(x), \dots$

§ Generating Function for $P_n(x)$

$$(1-2xz+z^2)^{-\frac{1}{2}} = \sum_{n=0}^{\infty} z^n P_n(x) \rightarrow \textcircled{B}$$

Expanding the LHS and equating the co-effs of the similar terms, we can find Legendre's polynomials $P_0(x), P_1(x), P_2(x), \dots$

[PS. Problems related to Legendre's polynomials can be solved easily by relations $\textcircled{A}$ & $\textcircled{B}$]

Ex Establish the results

(i) $P_0(x) = 1$

(ii) $P_1(x) = x$

(iii) $P_2(x) = \frac{1}{2} (3x^2 - 1)$

(iv) $P_3(x) = \frac{1}{2} (5x^3 - 3x)$

(v) $P_4(x) = \frac{1}{8} (35x^4 - 30x^2 + 3)$

(1) from (B), we write

$$(1 - 2xz + z^2)^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(x)$$

$$\Rightarrow [1 - (2xz - z^2)]^{-1/2} = \sum_{n=0}^{\infty} z^n P_n(x) \quad (V)$$

$$\Rightarrow 1 + \frac{1}{2}(2xz - z^2) + \dots = P_0(x) + z P_1(x) + z^2 P_2(x) + \dots$$

$$\Rightarrow \boxed{P_0(x) = 1}$$

(1 $\rightarrow$ const in the LHS)
($P_0(x) \rightarrow$ ----- RHS)

(ii) from (A) we write

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Put $n=1$, $P_1(x) = \frac{1}{2 \cdot 1!} \frac{d}{dx} (x^2 - 1)^1$

$$\Rightarrow P_1(x) = \frac{1}{2} \frac{d}{dx} (x^2 - 1)$$

$$\Rightarrow \boxed{P_1(x) = x}$$

(iii) $n=2$, $P_2(x) = \frac{1}{2^2 \cdot 2!} \frac{d^2}{dx^2} (x^2 - 1)^2$

$$= \frac{1}{8} \frac{d}{dx} \left[\frac{d}{dx} \{ x^4 - 2x^2 - 1 \} \right]$$

$$= \frac{1}{8} \frac{d}{dx} (4x^3 - 4x)$$

$$= \frac{1}{8} (12x^2 - 4)$$

$$\boxed{P_2(x) = \frac{1}{2} (3x^2 - 1)}$$

(4)

$$(iv) P_3(x) = \frac{1}{2^3 \cdot 3!} \frac{d^3}{dx^3} (x^2 - 1)^3 \quad (\text{Put } n=3 \text{ in } \textcircled{A})$$

$$P_3(x) = \frac{1}{2} (5x^3 - 3x) \quad \underline{HW}$$

$$(v) P_4(x) = \text{Put } n=4 \text{ in } \textcircled{A} \text{ and differentiate} \quad \underline{HW}$$

$$(vi) P_5(x) = \frac{1}{8} (63x^5 - 70x^3 + 15x) \quad \underline{HW}$$

$$(vii) P_6(x) = ? \quad \underline{HW}$$

Important to remember

(i) Legendre's Eqⁿ

(ii) Legendre's Polynomial, if possible
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(iii) Rodrigue's Formula (most important)
 $\textcircled{A}$

(iv) Generating Function
 $\textcircled{B}$